

USING PARETO AND WEIBULL DISTRIBUTIONS IN THE MODELLING OF GROWTH PROCESSES

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Key words: Dislocations; fluidized bed; galaxies; Pareto distribution; silicon wafer; size distribution; truncation; Weibull distribution; wildfire.

Summary: Deterministic growth processes with random stopping times are presented which can lead to Pareto and Weibull distributions and truncated forms of these distributions. These models are used in statistical analyses of various phenomena such as wildfire sizes, numbers of dislocations on silicon wafers, numbers of galaxies in cubic cells of the universe and size of particles in a process of chemical engineering.

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1. Introduction

The Pareto and Weibull distributions are classical probability distributions, which are frequently used in the statistical literature. The Pareto distribution is popular as a heavy tail distribution and has been used in the last years, for example, in modelling rainfalls (Salvadori and De Michele, 2001; Coles and Pericchi, 2003), the sizes of earthquakes (Kagan and Schoenberg, 2001) and wildfires (Reed and McKelvey, 2002; Schoenberg et al., 2003). The Weibull distribution is a popular lifetime distribution in reliability theory (Lee and Engelhardt, 1991; Bracquemond and Gaudoin, 2003), an important particle size distribution (Fang, et al., 1993, in this context also called Rosin-Rammler distribution, Brown and Wohletz, 1995) and a material strength distribution (Lindquist, 1994; Lu et al., 2002). Usually the shape parameter κ of the Weibull distribution in (5) is taken as $\kappa > 1$.

Obviously, in many cases the Pareto distribution is used only as a formal approximation, while for the Weibull distribution various models exist. The aim of this paper is to show that both distributions result from simple stochastic models, which may explain the success of their use and help to interpret statistically estimated model parameters. It is interesting that in the context of these models the Weibull shape parameter κ is often smaller than 1, which is unusual in the classical applications. The models are simple growth models which are based on natural deterministic growth models and get their random nature by random stopping times. In the spirit of this approach many other models are possible.

The application of the models is demonstrated in statistical analyses of wildfire and particle sizes and dislocation and galaxies numbers.

The present paper is in the spirit of Reed and McKelvey (2002), who describe such models in terms of growth and extinguishment rates and apply them in the statistical analysis of wild fires and many other cases different to the cases studied here. However, those authors do not consider Gumbel and truncated Pareto and Weibull distributions.

2. Statistical distributions

2.1 Pareto distribution

A random variable X with the distribution function

$$F(x) = 1 - \left(\frac{a}{x}\right)^\lambda \quad \text{for } x \geq a \text{ and } \lambda, a > 0, \quad (1)$$

is said to have a Pareto distribution with parameters a and λ ; see Johnson et al. (1995). Its density function has the form

$$f(x) = \lambda a^\lambda x^{-(1+\lambda)}, \quad x \geq a \text{ and } \lambda, a > 0.$$

The mean and variance of X are

$$E(X) = \frac{\lambda a}{\lambda - 1} \quad \text{for } \lambda > 1,$$

and

$$Var(X) = \frac{\lambda a^2}{(\lambda - 1)^2(\lambda - 2)} \quad \text{for } \lambda > 2.$$

For $\lambda \leq 1$ the mean of X does not exist.

If T has an exponential distribution with mean $1/\lambda$, $T \sim \text{Exp}(\lambda)$, then the random variable

$$X = ae^T \quad (2)$$

has a Pareto distribution with parameters a and λ .

Modifications of the Pareto distribution are the tapered and truncated Pareto distributions, see Johnson et al. (1995).

The tapered Pareto distribution has the form

$$F(x) = 1 - \left(\frac{a}{x}\right)^\lambda \exp\left(\frac{a-x}{b}\right) \quad \text{for } x > a \text{ and } b > 0. \quad (3)$$

For truncated Pareto see below in section 3.

The conditional distribution of a Pareto distributed random variable X in given $X > b$ with $b > a$ is

$$P(X < x | X > b) = 1 - \left(\frac{b}{x}\right)^\lambda \quad \text{for } x \geq b \text{ and } \lambda > 0, \quad (4)$$

i.e. it is a Pareto distribution with parameters b and λ .

2.2 Weibull distribution

A random variable X with the distribution function

$$F(x) = 1 - e^{-\lambda x^\kappa} \quad \text{for } x \geq 0 \text{ and } \lambda, \kappa > 0, \quad (5)$$

is said to have a Weibull distribution with parameters λ and κ . For the use in this paper, we introduce the real-valued parameter m by $\kappa = 1/m$ and write the Weibull distribution as

$$F(x) = 1 - e^{-\lambda x^{\frac{1}{m}}} \quad \text{for } x \geq 0 \text{ and } m, \lambda > 0. \quad (6)$$

Its density function has the form

$$f(x) = \frac{\lambda}{m} x^{\frac{1}{m}-1} e^{-\lambda x^{\frac{1}{m}}} \quad \text{for } x \geq 0 \text{ and } m, \lambda > 0.$$

(Note that the density function has a pole at $x = 0$ for $m > 1$.)

The mean and variance of X are

$$E(X) = \left(\frac{1}{\lambda}\right)^m \Gamma(1+m),$$

and

$$\text{Var}(X) = \left(\frac{1}{\lambda}\right)^{2m} (\Gamma(1+2m) - \Gamma^2(1+m)),$$

where $\Gamma(\cdot)$ denotes the gamma function.

If $T \sim \text{Exp}(\lambda)$ then the random variable

$$X = T^m \quad (7)$$

has a Weibull distribution with parameters λ and $1/m$.

The conditional distribution of X given $X > a$ is

$$P(X < x | X > a) = 1 - e^{-\lambda(\sqrt[m]{x} - \sqrt[m]{a})} \quad \text{for } x \geq a \text{ and } m, \lambda > 0. \quad (8)$$

Note that this is not the so-called three-parameter Weibull distribution.

2.3 Gumbel distribution

A random variable T with distribution function

$$F(t) = 1 - \exp(-e^{\frac{t-\alpha}{\beta}}) \quad \text{for } -\infty < t, \alpha < +\infty, \beta > 0, \quad (9)$$

is said to have a Gumbel distribution with parameters α and β . Its density function has the form

$$f(t) = \frac{1}{\beta} e^{\frac{t-\alpha}{\beta}} \exp(-e^{\frac{t-\alpha}{\beta}}) \quad \text{for } -\infty < t, \alpha < \infty, \beta > 0.$$

The mean and variance of T are

$$E(T) = \alpha - 0.577216\beta,$$

and

$$\text{Var}(T) = \frac{\pi^2}{6} \beta^2.$$

If $T \sim \text{Gum}(-\frac{m}{\mu} \log \lambda, \frac{m}{\mu})$ then the random variable

$$X = e^{\mu T}, \quad (10)$$

has a Weibull distribution with parameters λ and $1/m$.

3. Stochastic growth models leading to Pareto and Weibull distributions

In the following, we reinterpret the equations (2), (7) and (10) as stochastic growth models, which lead to the Pareto or Weibull distributions. Consider a random variable $X(T)$ which is constructed as follows.

There is a deterministic law of growth $X(t)$ according to which some size or number variable increases in time t . This growth process is stopped at a random time T so that the growth process ends with the value $X(T)$. Thus $X(T)$ is simply a function of the random variable T . The value of this approach of modelling lies in the possibility of physical interpretation of the parameters of the various distributions.

Table 1 shows some particular cases which are of interest for the present article.

Table 1. Growth processes with different distributions of random stopping times, which lead to Pareto (truncated Pareto) and Weibull (truncated Weibull) distributions.

Growth law	Distribution of T	Resulting distribution
$ae^{\mu t}$	$\text{Exp}(\lambda)$	Pareto $\left(a, \frac{\lambda}{\mu}\right)$
$e^{\mu t}$	$\text{Gumbel}\left(-\frac{m}{\mu} \log \lambda, \frac{m}{\mu}\right)$	Weibull $\left(\lambda, \frac{1}{m}\right)$
t^m	$\text{Exp}(\lambda)$	Weibull $\left(\lambda, \frac{1}{m}\right)$
$ae^{\mu t}$	one-side trunc. $\text{Exp}(\lambda)$	one-side trunc. Pareto $\left(a, \frac{\lambda}{\mu}\right)$
t^m	one-side trunc. $\text{Exp}(\lambda)$	one-side trunc. Weib.
t^m	two-sides trunc. $\text{Exp}(\lambda)$	two-sides trunc. Weib.

In the following, two of the growth processes in Table 1 which lead to truncated Pareto and Weibull distributions are discussed in detail. Consider the growth process

$$X(t) = ae^{\mu t} \quad \text{for } a > 0. \quad (11)$$

It is stopped at a random time T which has an exponential distribution with mean $1/\lambda$ truncated-from-above (at $\log \frac{b}{a}$) with distribution function

$$F^{\log \frac{b}{a}}(t) = \frac{1 - e^{-\lambda t}}{1 - e^{-\lambda \log \frac{b}{a}}} \quad \text{for } 0 < t < \log \frac{b}{a} \text{ and } b > a. \quad (12)$$

Then $X(T)$ has a Pareto distribution with parameters a and $\frac{\lambda}{\mu}$ truncated-from-above at b , with distribution function

$$G_a^b(x) = \frac{1 - (\frac{a}{x})^{\frac{\lambda}{\mu}}}{1 - (\frac{a}{b})^{\frac{\lambda}{\mu}}} \quad \text{for } a < x < b. \quad (13)$$

Furthermore, if the growth process

$$X(t) = t^m \quad \text{for } m > 0, \quad (14)$$

is stopped at a random time T which has an exponential distribution with mean $1/\lambda$ truncated-from-below at $\sqrt[m]{a}$ with distribution function

$$F_{\sqrt[m]{a}} = 1 - e^{(-\lambda(t - \sqrt[m]{a}))} \quad \text{for } t > \sqrt[m]{a}, \quad (15)$$

then $X(T)$ has a Weibull distribution with parameters λ and $1/m$ truncated-from-below at a given in (8). The situation in the case of truncation from above is analogous. By the way, for continuous monotone $X(t)$ it is true in general that truncated stopping time distributions lead to truncated distributions of $X(T)$ because $X(T)$ is monotone in T .

Depending on the problem under study different forms for the growth process $X(t)$ and random stopping times T might be considered, which yield different distribution functions of $X(T)$. These distributions may have complicated forms and may not yet have appeared in the theory of statistical distributions. Consider, for example, the growth process

$$X(t) = (\sqrt[m]{a} + t)^m \quad \text{for } t > 0.$$

If the random stopping time T has a gamma distribution with parameters α and λ and probability density function

$$f(t) = \frac{\lambda^\alpha}{\Gamma(\alpha)} t^{\alpha-1} \exp(-\lambda t) \quad \text{for } t > 0 \text{ and } \alpha, \lambda > 0, \quad (16)$$

then $X(T)$ has the probability density function

$$f(x) = \frac{1}{m} x^{\frac{1}{m}-1} \frac{\lambda^\alpha}{\Gamma(\alpha)} (\sqrt[m]{x} - \sqrt[m]{a})^{\alpha-1} \exp(-\lambda(\sqrt[m]{x} - \sqrt[m]{a})) \quad \text{for } x > a. \quad (17)$$

In the following the reader should note that classes of continuous distribution functions are examined as distributions of random integers, while integers are of course discrete. However, for the large considered integers this discrepancy does not matter.

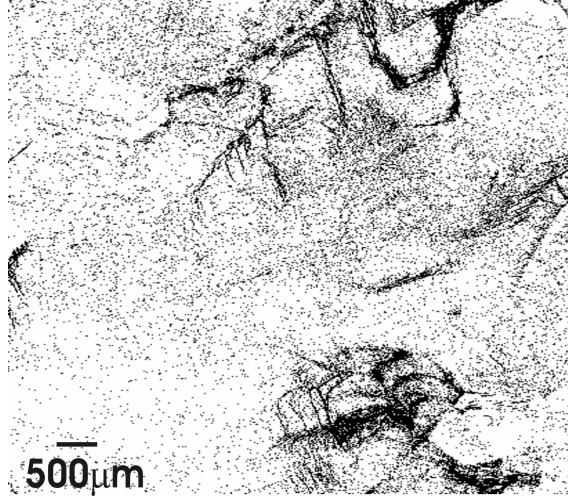


Figure 1. A typical sample of a dislocation pattern on a mc-si wafer. The darker regions correspond to dislocation clusters of very high densities. By courtesy of Markus Rinio.

4. Examples

4.1 Dislocation number distribution

4.1.1 The data

In multicrystalline silicon (mc-si) crystals line defects appear, which are called dislocations. Dislocations as well as other defects in crystals (such as grain boundaries and microprecipitate) control the electrical properties of the material and often impact the solar cell characteristics adversely. A volume of 1 mm^3 may contain dislocations of a total length of 1 km.

For statistical analysis of dislocations on silicon wafers, which represent planar sections through the crystal, samples were provided on which the dislocations appear as small dots. Figure 1 shows a microscopic image of a small piece of a wafer, which is in general an extremely irregular point pattern and shows a high degree of clustering. It does not make sense to model them as samples of stationary point processes and the obvious anisotropies do not show a pattern which encourages statistical modelling. Thus we introduced a new summary statistic, which compresses the variability very much but enables a convenient comparison of different samples.

The point pattern on the wafer is considered as an inhomogeneous planar point process with intensity function $\lambda(x)$. Define the *point density distribution function* as

$$D(t) = \lim_{b \rightarrow \infty} \frac{\nu(R_t)}{b^2}, \quad (18)$$

where $\nu(\cdot)$ denotes the Lebesgue measure and R_t is the set of all points in the square $[0, b]^2$ of side-length b with $\lambda(x) \leq t$,

$$R_t = \{x \in \mathbb{R}^2 : \lambda(x) \leq t\}.$$

This function can be estimated statistically by

$$\hat{D}(t) = \frac{\text{area with estimated point density } \hat{\lambda}(x) \leq t}{\text{area of } W} = \frac{\nu(W_t)}{\nu(W)}. \quad (19)$$

Here W is the window of observation and W_t the subset of points x satisfying $\hat{\lambda}(x) \leq t$,

$$W_t = \{x \in W : \hat{\lambda}(x) \leq t\},$$

where $\hat{\lambda}(x)$ is an estimate of the point density at point x .

Another way for defining the point density distribution function is as follows. Suppose that $\{\lambda(x)\}$ is a stationary random field on $\mathbb{R}^2$, which is observed in the window W . The point process could be in this case a Cox process (Stoyan et al., 1995) with random intensity $\{\lambda(x)\}$. Then $D(t)$ in (18) is the one-dimensional marginal distribution of the random field.

Figure 2 shows the fluctuations of the decadic logarithm of the intensity function for the pattern of Figure 1, which was estimated by physicists (Rinio et al., 2002); in the black region the dislocation density is less than 10^5 cm^{-2} . The fraction of the area of the black region is $\hat{p} = 0.68$. Figure 3 shows the corresponding empirical point density distribution function.

4.1.2 A physical model for dislocation growth

The growth process of dislocations in mc-si is well studied in the physical literature. Physicists speak about dislocation multiplication. The so-called multiplication behaviour of the number $X(t)$ of dislocations at time t in a fixed area is described by the differential equation

$$\frac{dX(t)}{dt} = K \frac{v_0}{\tau_0^m} \exp\left(\frac{-U}{kT}\right) \left(\tau_a - A\sqrt{X(t)}\right)^{m+1} X(t) \quad (20)$$

with initial condition $X(0) = 1$, where $k, v_0, \tau_0, \dots$ are physical and material parameters, see Franke (2000) and Franke et al. (2000).

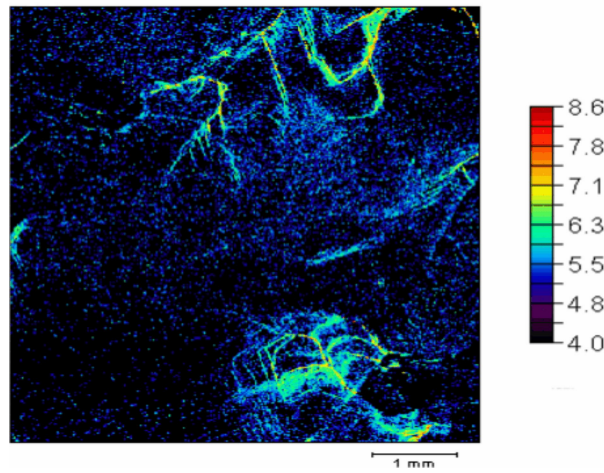


Figure 2. Fluctuations of $\lg \hat{\lambda}(x)$ in Fig. (1). The unit of intensities is cm^{-2} .

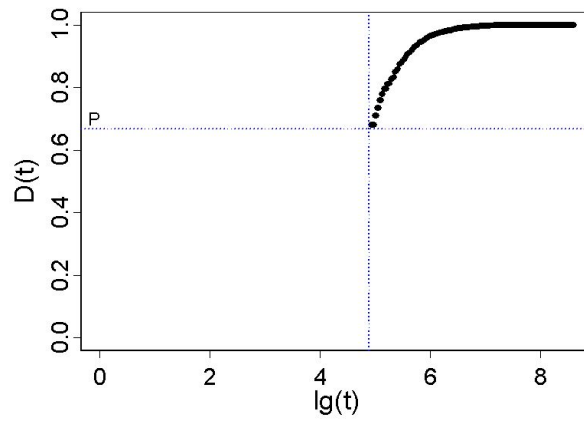


Figure 3. The empirical point density distribution function of the sample in Fig. 2, which is estimated using formula (19). It is $\hat{p} = 0.68$. For $t < 10^5 \text{ cm}^{-2}$ no estimates of $D(t)$ are given.

For the given material (Figures 1, 2) there is clear evidence that the physical process of dislocation growth acts under conditions where $A = 0$ can be assumed (details in Ghorbani, 2004). In this case the solution of differential (20) equation is

$$X(t) = e^{\mu t}, \quad (21)$$

where $\mu = K \frac{v_0}{\tau_0^m} \exp(\frac{-U}{kT}) \tau_a^{m+1}$, which equals 0.042 sec^{-1} for the sample considered, see Ghorbani (2004).

Thus for a given growth time t the number of dislocations seems to be calculable. However, as Figure 2 clearly shows, the density of dislocations varies heavily. Thus the deterministic law (21) describes only some mean behaviour of the growth process, and for modelling the fluctuations a stochastic model is necessary. Here the approach with deterministic growth and random stopping time is used, assuming that the variability of $\hat{\lambda}(x)$ is caused by local variability of growth time.

4.1.3 Fitting distributions to the empirical point density distribution function

Consider the exponential growth process of dislocations

$$X(t) = e^{\mu t}, \quad (22)$$

and assume then that the growth process is stopped at a random time T . The authors tried first an exponential distribution with parameter λ , believing that for this assumption there is some statistical support, see Ghorbani (2004). In this case, $X(T) = e^{\mu T}$ follows a Pareto distribution with parameters $a = 1$ and $\theta = \frac{\lambda}{\mu}$.

The needed parameter θ was estimated by fitting the Pareto distribution with parameters $a = 1$ and θ to the estimated censored point density

distribution function in Figure 3 using the least squares method. (Here and at other places in this paper the nature of the given data enforce the application of the least squares method, as only the empirical distribution function is given and not a sample of realizations.) This yielded $\hat{\theta} = 0.19$, i.e. $\lambda = 0.042 \times 0.19 = 0.008$. The corresponding mean growth time is estimated as 125 seconds.

However, Figure 4 shows that the obtained Pareto distribution does not fit the data. Furthermore, since $\theta < 1$, $X(T)$ has an infinite mean, which is physically not plausible. Consequently, the assumption of an exponentially distributed growth time seems to be false.

An alternative model for the stopping time is based on the fact that the Weibull distribution provides an excellent fit to the empirical data, see Figure 4.

Since the unique distribution for the random growth time T which yields in the case of exponential growth the Weibull distribution with parameters λ and $1/m$ for $X(T)$ is a Gumbel distribution with parameters $\alpha = -\frac{m}{\mu} \log \lambda$ and $\beta = \frac{m}{\mu}$, this distribution was used for the random stopping time.

The needed parameters are estimated as $\alpha = 257.96$ and $\beta = 54.76$, which were obtained using the estimated parameters of the Weibull distribution and the physical parameter $\mu = 0.042$ given in (21). The corresponding mean growth time is estimated as 226 seconds. It must be mentioned that the random variable T is a time, which must be positive. But according to (9) the Gumbel distributed random variable T also can get negative values with probability 0.009. Fortunately, this probability is very small. Summarizing, the original deterministic physical theory (20) was refined to a plausible stochastic model.

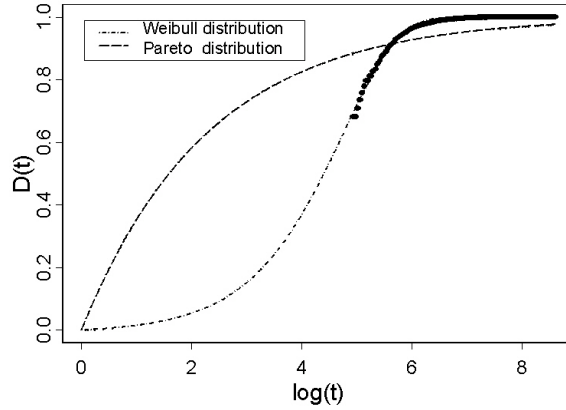


Figure 4. Fit of the Pareto and Weibull distributions to the empirical point density distribution function (•) for the sample in Fig. 2.

4.2 Distribution of the Number of Galaxies

Modern astronomers study extensively the distribution of galaxies in the universe. This includes also statistical analysis of numbers of galaxies in cells (cell counts), since they are easier to deal with than the raw, unbinned data. There exist various models for cell count distributions, see Martínez and Saar (2002). Classical point process statistics would probably suggest for such counts the negative binomial distribution because of the high degree of clustering of galaxies. This model has been used also by cosmologists, see Elizalde and Gaztañaga (1992) and Betancort-Rijo (2000).

The data analyzed here are simulated numbers of galaxies in a cubic window of the universe, provided to the authors by Prof. V.J. Martínez, using N-body cosmological simulation. It is based on a complicated dynamic model

for the development of the universe, see Bagla (2005), which produces samples astronomers consider as worthy for statistical analysis. The original window with side length of 141.3 Mpc (1 Mpc is a million of parsecs, 1 parsec is 3.26 light years) was divided into 27000(=30³) subwindows, called cells, and the numbers of galaxies were counted. The empirical mean and variance of the cell counts are 0.57 and 1.98, respectively.

We started the statistical analysis with negative binomial distribution, which is given by the probability mass function

$$P(X = k) = \frac{\Gamma(\theta + k)a^\theta}{k! \Gamma(\theta)(1 + a)^{(\theta+k)}} \quad \text{for } k = 0, 1, 2, \dots \text{ and } a, \theta > 0. \quad (23)$$

As an alternative in the spirit of this paper, also the Weibull distribution was fitted to the data. This approach seems to be appropriate as there are some models for the galaxy clustering process that are based on the fact that the universe is seen by cosmologists as a result of a growth process, see Bernard et al. (2004). However, as for many of our cells the number of galaxies is small, we do not work with the continuous Weibull distribution but instead by its discretized version with probability mass function

$$P(X = k) = e^{-\lambda k^{\frac{1}{m}}} - e^{-\lambda(k+1)^{\frac{1}{m}}} \quad \text{for } k = 0, 1, 2, \dots \text{ and } \lambda, m > 0. \quad (24)$$

The so-called discrete Weibull distribution given by (24) is not unknown in the statistical literature (Nakagawa and Osaki, 1975; Johnson et al., 1992) but was probably never before used in cosmology.

Table 2 shows the estimated parameters of models in (23) and (24) which were obtained by the maximum likelihood method using the Newton-Raphson optimization procedure, see Venables and Ripley (2002). The standard errors (SEs) given in parentheses were obtained via the diagonal of the inverse of

the Hessian matrix of the log-likelihood function. The theoretical mean and variance of number of galaxies per cell for two models are 0.57 and 1.76, respectively.

Table 2. Maximum likelihood estimates of the parameters of the and the negative binomial and discrete Weibull distributions for the galaxy data. The corresponding AICs of the models are 52460.18 and 52406.58 respectively.

Model	$\lambda(\text{SE})$	$\hat{m}(\text{SE})$	$\theta(\text{SE})$	$\hat{a}(\text{SE})$
Disc. Weib.	1.32(0.010)	1.55(0.014)	—	—
Neg. bin.	—	—	0.28(0.006)	0.49(0.012)

To compare the goodness-of-fit of the two models the Akaike Information Criterion (AIC) is used. The AIC is a likelihood-based statistic and includes a penalty for each parameter in the model (Akaike, 1983) and is computed as twice the negative likelihood plus twice the number of estimated parameters; a lower AIC indicates a better fit.

Comparison of the AICs of the two models given in Table 2 shows that the discrete Weibull fits the data better than the negative binomial distribution. The empirical and theoretical numbers are shown in Table 3.

Serfozo (1990) introduced two point processes in which the number of points in disjoint cells has a negative binomial distribution, see also Diggle and Milne (1983). But probably until now no point processes have been introduced in which the number of points in disjoint cells has a discrete Weibull distribution. The question remains open to interpret the parameters λ and m cosmologically.

Table 3. The empirical frequencies of numbers of the galaxies and the corresponding theoretical values of the negative binomial distribution in (23) and the discrete Weibull distribution in (24).

Numbers	Empirical numbers	Weibull	Neg. bin.
0	19755	19780.60	19794.94
1	3842	3796.15	3697.23
2	1618	1580.40	1590.30
3	756	778.56	813.04
4	417	420.08	448.64
5	222	240.58	258.48
6	116	143.84	153.12
7	89	88.89	92.47
8	54	56.41	56.65
9	35	36.59	35.09
10	30	24.17	21.92
11	16	16.23	13.79
12	12	11.04	8.73
13	11	7.61	5.55
14	7	5.30	3.54
15	3	3.73	2.27
16	2	2.64	1.46
17	2	1.89	0.94
18	5	1.36	0.60
19	1	0.99	0.39
20	1	0.72	0.25
21	0	0.53	0.16
22	1	0.39	0.10
23	1	0.29	0.07
24	1	0.21	0.04
25	0	0.16	0.03
26	0	0.12	0.01
27	0	0.09	0.01
28	2	0.07	0.00
29	1	0.05	0.00

4.3 Wildfire size distributions

In Schoenberg et al. (2003) a variety of statistical distributions were examined for the Los Angeles County wildfire sizes between 1950 and 2000 greater than 0.15625 square miles. The AIC likelihood criterion and graphical methods as well as Kolmogorov-Smirnov and Cramér-von Mises statistics were used to compare the goodness-of-fits of the different distributions. The result was that the truncated-from-above Pareto distribution yields a better fit than the other alternative distributions like Pareto, tapered Pareto etc.

Schoenberg et al. (2003) did not offer any model for the wildfire growth process. The growth models of this paper can easily be used (see also Reed and McKelvey, 2002).

Suppose that $X(t)$ represents the wildfire size at time t , which follows the deterministic model $X(t) = ae^{\mu t}$ with $X(0) = a = 0.15625$ square miles. Assume that a wildfire is stopped at a random time T , which can be interpreted as lifetime of fire. If T has an exponential distribution with mean $1/\lambda$ truncated-from-above at $\log \frac{b}{a}$ with distribution function given in (12) then $X(T)$ has a Pareto distribution with parameters a and $\frac{\lambda}{\mu}$ truncated-from-above at b , with distribution function given in (13).

We tried to fit also the truncated-from-below Weibull distribution in (8) to Schoenberg's data, but this did not lead to a better fit than with the truncated Pareto distribution since the corresponding AIC of these models are 1440 and 1424, respectively.

4.4 Particle size distribution in the fluidized bed

In chemical engineering there are several processes where agglomeration is used to combine smaller particles in order to get larger particles. The final

product is a coarse grained material which does not agglomerate, is easy to store and move and easily mixes with other materials.

One such process is the fluidized bed spray granulation method, see Pietsch (2002). The preexisting solid particles in the fluidized bed are sprayed with a liquid from the top down or bottom up. The solvent evaporates in the hot unsaturated gas and the remaining solid matter grows in layers on the particle surface. Growth through agglomeration of particles with each other also occurs. If the particles achieve a particular diameter then they are removed from the apparatus.

In the following a simple stochastic model for the random particle growth process is discussed, which is in the spirit of the present paper but leaves the field of Weibull or Pareto distributions.

Let $X(t)$ be the diameter of a particle at time t . Heinrich (2002) showed that $X(t)$ grows in process as described above theoretically according to the deterministic law

$$X(t) = X(0) \sqrt[3]{\frac{1}{6}t + 1}, \quad (25)$$

in which the unit of time is $[h]$. However, the process of particle growth is to some extent random, as Figure 5 shows for some particular experiments.

In contrast to the models in the sections above, the starting value $X(0)$ of the growth process is here also random. For the experiment considered it was known that $X(0)$, the particle diameter in the beginning, has a normal distribution with parameters $\mu_0 = 869.87 \mu m$, $\sigma_0 = 108.15 \mu m$. At the end, when the particle leaves the apparatus, the mean and the standard deviation are $\mu = 1278.95 \mu m$ and $\sigma = 276.89 \mu m$. The difference cannot only be explained by the growth law (25). Thus it is assumed that the random

residence time of the particles in the fluidized bed also plays a role. This time is interpreted as a stopping time as in the other models of this paper.

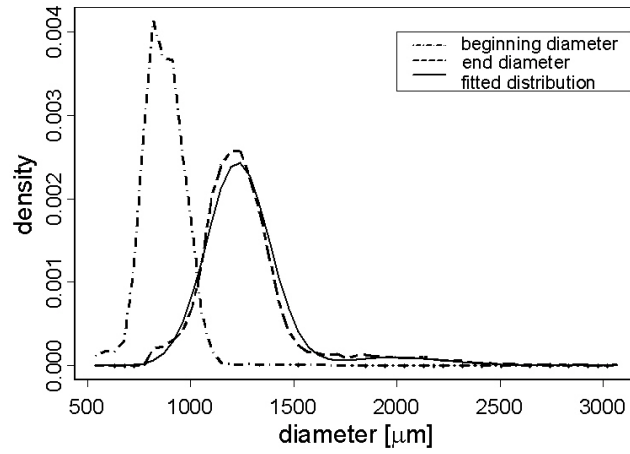


Figure 5. The empirical density function of particle diameter $X(0)$ in the beginning and $X(T)$ at the end and the fitted distribution to $X(T)$ for an experiment.

In chemical engineering it is known that such residence times often follow exponential or gamma distributions. The authors tried such distributions without success and finally came to a mixture of two gamma distributions with probability density function

$$f(t) = \begin{cases} \Gamma(\alpha_1, \lambda_1)(t) & \text{with probability } p, \\ \Gamma(\alpha_2, \lambda_2)(t) & \text{with probability } 1 - p, \end{cases}$$

($\Gamma(\alpha_i, \lambda_i)(t)$ denotes the probability density function of the gamma distribution with parameters α_i and λ_i as given in (16)).

The parameters p , α_i and λ_i were estimated so that the distribution of $X(T)$ with normally distributed $X(0)$ is close to the empirical distribution

in Figure 5. This yielded $\hat{p} = 0.937$, $\hat{\alpha}_1 = 96.43$, $\hat{\lambda}_1 = 8.61$, $\hat{\alpha}_2 = 3467.50$ and $\hat{\lambda}_2 = 51.60$. The corresponding mean and standard deviation of particle diameter at the end are $1278.51 \mu m$ and $246.77 \mu m$.

The results show that there are two components of the residence time distribution, a short with mean 11.2 and a long with mean 67.2, which correspond to small and large particles.

The estimation method assumes that $F_0(\cdot)$ and $F(\cdot)$ are the distribution functions of $X(0)$ and $X(T)$ and $f_T(\cdot)$ is the density function of the residence time T . Reexpress equation (25) as

$$X(T) = X(0)g(T).$$

This yields

$$\begin{aligned} F(x) &= P(X(T) \leq x) = P(X(0) \leq \frac{x}{g(T)}) \\ &= \int_0^\infty F_0\left(\frac{x}{g(t)}\right) f_T(t) dt. \end{aligned}$$

Least squares estimation can be accomplished by the following steps. At first, the density function of T is parameterized as $f_T(\theta, \cdot)$, then the parameters of $f_T(\theta, \cdot)$ are estimated by minimizing the following sum of square errors:

$$\sum_{i=1}^n \left(\hat{F}(x_i) - \int_0^\infty \hat{F}_0\left(\frac{x_i}{g(t)}\right) f_T(\theta, t) dt \right)^2,$$

for some values x_i of particle diameter and the empirical values of $F_0(\cdot)$ and $F(\cdot)$.

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References

- AKAIKE, H. (1983). Information measure and model selection. *Bull. Internat. Stat. Inst.*, **50**:277–291.
- BAGLA, J.S. (2005). Cosmological N-body simulation: Techniques, scope and status. *Curr. Sci.*, **88**:1088–1100.
- BERNARD, J.T.J., MARTÍNEZ, V.J., SAAR, E. & TRIMBLE, V. (2004). Scaling laws in the distribution of galaxies. *Rev. Mod. Phys.*, **76**:1211–1266.
- BETANCORT-RIJO, J. (2000). Generalized negative Binomial distributions. *J. Stat. Phys.*, **98**:917–933.
- BRACQUEMOND, C. & GAUDOIN, O. (2003). A survey on discrete lifetime distributions. *International Journal of Reliability, Quality and Safety Engineering*, **10**(1):69–98.
- BROWN, W.K. & WOHLTZ, K.H. (1995). Derivation of the Weibull distribution based on physical principles and its connection to the Rossin-Rammler and lognormal distributions. *J. Appl. Phys.*, **78**(4):2758–2763.
- BRYANT, G., MARTIN, S., BUDI, A., & VAN MEGEN, W. (2003). Accurate measurement of small polydispersities in colloidal suspensions. *Langmuir*, **19**(3):616–621.
- COLES, S. & PEROCCHI, L. (2003). Anticipating catastrophes through extreme value modelling. *J. Roy. Statist. Soc. Ser. C*, **52**(4):405–416.
- DIGGLE, P.J. & MILNE, R.K. (1983). Bivariate Cox processes: some models for bivariate spatial point patterns. *J. R. Stat. Soc. Ser. B, Stat. Methodol.*, **45**(1):11–21.

- ELIZALDE, E. & GAZTAÑAGA, E. (1992). Void probability as a function of the void's shape and scale-invariant models. *Mon. Not. R. Astron. Soc.*, **254**:247–256.
- FANG, Z.G., PATTERSON, B.R. & TURNER, M.E. (1993). Modelling particle-size distributions by the Weibull distribution function. *Mater. Charact.*, **31**(3):177–182.
- FRANKE, D. (2000). *Numerische Simulation der Versetzungsmultiplikation in multikristallinem Silicium aus der gerichteten Blockkristallisation*. Ph.D. Thesis, RWTH Aachen Universität.
- FRANKE, D., APEL, M., HÄSSLER, C. & KOCH, W. (2000). Effect of multicrystalline structure on dislocation multiplication in cast Silicon. In *16th European Photovoltaic Solar Energy Conference*, pages 1317–1320, Glasgow, UK.
- GHORBANI, H. (2004). *Methods of Spatial Statistics for the Characterization of Dislocation Systems*. Ph.D. Thesis, Technische Universität Bergakademie Freiberg.
- HEINRICH, S. (2000). *Modellierung des Wärme und Stoffübergangs sowie der Partikelpopulationen bei der Wirbelschicht-Sprühgranulation*. Ph.D. Thesis, Otto-von-Guericke-Universität Magdeburg.
- JOHNSON, N.L., KOTZ, S. & KEMP, A.W. (1992). *Univariate Discrete Distributions*. J. Wiley & Sons, New York, Chichester, Brisbane, Toronto, Singapore.
- JOHNSON, N.L., KOTZ, S. & KEMP, A.W. (1995). *Continuous Univariate Distributions*. J. Wiley & Sons, New York, Chichester, Brisbane, Toronto, Singapore.
- KAGAN, Y. & SCHOENBERG, F. (2001). Estimation of the upper cutoff parameter for the tapered Pareto distribution. *J. Appl. Prob.*, **30A**:901–918.
- LEE, J.B. & ENGELHARDT, M. (1991). *Statistical Analysis of Reliability and Life-Testing Models: Theory and Methods*. Marcel Dekker, New York.
- LINDQUIST, E.S. (1994). Strength of materials and the Weibull distribution. *Probabilist. Eng. Mech.*, **9**(3):191–194.

- LU, C., DANZER, R. & FISCHER, F.D. (2002). Fracture statistics of brittle materials: Weibull or normal distribution? *Phys. Rev.*, **E65**(6):1640–1642.
- MARTÍNEZ, V.J. & SAAR, E. (2002). *Statistics of the Galaxy Distribution*. Chapman & Hall/CRC, Boca Raton, London, New York, Washington, D.C.
- NAKAGAWA, T. & OSAKI, S. (1975). The discrete Weibull distribution. *IEEE Trans. Reliability*, **24**:330–301.
- PIETSCH, W. (2002). *Agglomeration Processes: Phenomena, Technologies, Equipment*. Wiley-VCH, Weinheim.
- REED W.J. & MCKELVEY, K.S. (2002). Power law behaviour and parametric models for the size-distribution of forest fires. *Ecological Modelling*, **150**:239–254.
- RINIO, M., PETERS, S., WERNER, M., LAWERENZ, A. & MÖLLER, H.J. (2002). Measurement of the normalized recombination strength of dislocations in multicrystalline Silicon solar cells. *Solid State Phen.*, **82–84**:701–706.
- SALVADORI, G. & DE MICHELE, C. (2001). From generalized Pareto to extreme values law: scaling properties and derived features. *J. Geophys. Res.*, **106**(D20):24063–24070.
- SCHOENBERG, F., PENG, R. & WOODS, J. (2003). On the distribution of wildfire sizes. *Environmetrics*, **14**(6):583–592.
- SERFOZO, R.F. (1990). Point processes. In Heyman, D.P. and Sobel, M.J., editors, *Handbook in Operations Research and Management Science: Stochastic Models*, pages 1–93. Dover Pubns.
- STOYAN, D., KENDALL, W. & MECKE, J. (1995). *Stochastic Geometry and Its Applications*. J. Wiley & Sons, Chichester.
- VENABLES, W.N. & RIPLEY, B.D. (2002). *Modern Applied Statistics With S*. Springer, New York, Berlin, Heidelberg.